article reproducing the letter and adding the following ‘practical’ reasons to curtail the scale of the operation in its list. [28]

CURIOUS DIFFICULTIES IN BUILDING RAILWAYS FOR INDIA: RAINS, INSECTS AND MORE

> The physical obstacles which are to be overcome or neutralized before a system of railways can be carried into execution in India, have, to a certain extent, been foreseen and pointed out by the East India directors, in their letter of the 7th of May last, addressed to the Governor General of India. In that letter, they observe, ‘Independent of the difficulties common to railroads in all countries, there are others peculiar to the climate and circumstances of India, which may render it advisable that the first attempt should be made upon a limited scale.
>
> These difficulties may be classed under the following heads:
>
> 1. Periodical rains and inundations
> 2. The continued action of violent winds and influence of a vertical sun
> 3. The ravages of insects and vermin upon timber and earthwork
> 4. The destructive effects of the spontaneous vegetation of underwood upon earth and brickwork
> 5. The unenclosed and unprotected tracts of country through which railroads would pass
> 6. The difficulty and expense of securing the services of competent and trustworthy engineers
>
> These obstacles, combined with the entire want of definite and scientific information relative to the applicability of railway communication to India, have induced the Company to take the step to which reference has been already made, that of employing a qualified engineer to survey the country, in order to fix upon the most eligible lines which might be available for the experiment.

PECULIARITIES FOR THE FIRST RAILWAY ENGINEERS

The *Railway Register* went on to add a few more ‘peculiarities’ of which the first railway engineers in India should be careful, which could turn cities to ruins. These were to be kept in mind by all those who attempted to carry out the work for the railway system in the country. Apart from the physical impediments that were mentioned, the following difficulties were also to be considered:

> First: The difficulties which the instability of soil itself presents
> Second: The impediments which the variations in the surface offer, these presenting features of a character wholly distinct from the ordinary difficulties which present themselves in climates and regions such as those of France, England and other European countries.